```
Arguments used:
  folder: C:\Users\pensf\Documents\AutomatedLBP_Benchmarking\LBP_Test_Images\Swatches\LBP_Texture_Swatches
  P: 8
  R: 1.0
  method: uniform
  metric: cosine
  save_csv: matches.csv
  blur: 0.0
  ltp: False
  visualize: True
  X: 64
  Y: 64
  V: True
  G: False
Correct matches: 38/75 (50.67%)
Highest distance among correct matches: 0.017436
Lowest distance among correct matches: 0.000408
Highest distance among incorrect matches: 0.003619
Lowest distance among incorrect matches: 0.000358
```

```
Index: 0
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000739
```

```
Index: 1
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.001657
```

```
Index: 2
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000589
```

```
Index: 3
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.000690
```

```
Index: 4
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.001447
```

```
Index: 5
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.001026
```

```
Index: 6
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Cosine
Distance: 0.002934
```

```
Index: 7
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.001463
```

```
Index: 8
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.003438
```

```
Index: 9
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Cosine
Distance: 0.002232
```

```
Index: 10
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.002521
```

```
Index: 11
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.001522
```

```
Index: 12
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.003370
```

```
Index: 13
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.001522
```

```
Index: 14
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000358
```

```
Index: 15
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000476
```

```
Index: 16
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000589
```

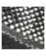
```
Index: 17
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.001725
```

```
Index: 18
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.001064
```

```
Index: 19
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.001064
```

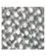
```
Index: 20
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.001568
```

```
Index: 21
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.004681
```



```
Index: 22
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.005825
```

```
Index: 23
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.001561
```

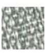


```
Index: 24
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.004058
```

```
Index: 25
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.001561
```

```
Index: 26
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.003619
```

```
Index: 27
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Cosine
Distance: 0.000739
```



```
Index: 28
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.001213
```

```
Index: 29
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.001074
```

```
Index: 30
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.000408
```

```
Index: 31
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.001904
```

```
Index: 32
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.000726
```

```
Index: 33
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.001624
```

```
Index: 34
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.000408
```

```
Index: 35
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.000690
```

```
Index: 36
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.001489
```

```
Index: 37
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000772
```

```
Index: 38
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Cosine
Distance: 0.001518
```

```
Index: 39
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.001074
```

```
Index: 40
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.001441
```

```
Index: 41
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.001218
```

```
Index: 42
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000476
```

```
Index: 43
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.001470
```

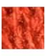
```
Index: 44
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.000358
```

```
Index: 45
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000549
```

```
Index: 46
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.001205
```

```
Index: 47
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000556
```

```
Index: 48
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.001661
```



```
Index: 49
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Cosine
Distance: 0.000624
```

```
Index: 50
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Cosine
Distance: 0.001205
```

```
Index: 51
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.001329
```

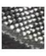
```
Index: 52
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Cosine
Distance: 0.001467
```

```
Index: 53
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.002420
```

```
Index: 54
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.002420
```

```
Index: 55
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Cosine
Distance: 0.002692
```

```
Index: 56
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.002693
```



```
Index: 57
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Cosine
Distance: 0.001568
```

```
Index: 58
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.005750
```

```
Index: 59
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.008020
```

```
Index: 60
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.005750
```

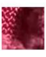
```
Index: 61
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.003639
```

```
Index: 62
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.003214
```

```
Index: 63
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Cosine
Distance: 0.003214
```

```
Index: 64
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.002937
```

```
Index: 65
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Cosine
Distance: 0.003872
```



```
Index: 66
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.001489
```

```
Index: 67
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.001133
```

```
Index: 68
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.001133
```

```
Index: 69
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Cosine
Distance: 0.001854
```

```
Index: 70
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Cosine
Distance: 0.001472
```

```
Index: 71
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.004977
```

```
Index: 72
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.001026
```

```
Index: 73
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.004977
```

```
Index: 74
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Cosine
Distance: 0.017436
```